

第三周作业

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1. 习题一:

24.

由题意得, 样本空间为男孩, 男孩, 男孩, 女孩, 女孩, 男孩, 女孩, 女孩

$$(1) P(A|\overline{B}) = \frac{P(A\overline{B})}{P(\overline{B})} = \frac{\frac{1}{4}}{\frac{1}{2}} = \frac{1}{2}$$

$$P(A|B) = \frac{P(AB)}{P(B)} = \frac{\frac{1}{4}}{\frac{1}{2}} = \frac{1}{2}$$

$$(2) \because P(AB) = \frac{1}{4}, P(A) = P(B) = \frac{1}{2}$$

$$\therefore P(AB) = P(A)P(B)$$

$\therefore A, B$ 相互独立

由 (1) 知 $P(A|\overline{B}) = P(A|B)$, B 是否发生对 A 没有影响, 故 A, B 相互独立。

25.

(1) 令 A = “两个人血型为 O 型, 其他三个人分别为其他三种血型”

由题意得不同人的血型是什么相互独立, 则

$$P(A) = C_5^2 A_3^3 0.46^2 \times 0.4 \times 0.11 \times 0.03 = 0.0168$$

(2) 令 B = “三个人为 O 型, 两个人为 A 型”

$$P(B) = C_5^3 0.46^3 \times 0.4^2 = 0.1557$$

(3) 令 C = “没有一人为 AB 型”

$$P(C) = (1 - 0.03)^5 = 0.8587$$

26.

令 $A_i (i = 1, 2, 3)$ 分别表示 “甲、乙、丙进球”, 则三个事件相互独立, 有

$$P(A_1) = 0.5, P(A_2) = 0.7, P(A_3) = 0.6 \quad (1) \text{ 令 } A = \text{“恰有一人进球”}$$

$$P(A) = P(A_1)\overline{P(A_2)}\overline{P(A_3)} + \overline{P(A_1)}P(A_2)\overline{P(A_3)} + \overline{P(A_1)}\overline{P(A_2)}P(A_3) = 0.5 \times (1 - 0.7) \times (1 - 0.6) + (1 -$$

$$2 \quad 0.5) \times 0.7 \times (1 - 0.6) + (1 - 0.5) \times (1 - 0.7) \times 0.6 = 0.29$$

(2) 令 B = “恰有两人进球”,

$$P(B) = P(A_1)P(A_2)P(\overline{A_3}) + P(\overline{A_1})P(A_2)P(A_3) + P(A_1)P(\overline{A_2})P(A_3) = 0.5 \times 0.7 \times (1 - 0.6) + (1 - 0.5) \times 0.7 \times 0.6 + 0.5 \times (1 - 0.7) \times 0.6 = 0.44$$

(3) 令 C = “至少有一人进球”

$$P(C) = 1 - P(\overline{C}) = 1 - P(\overline{A_1})P(\overline{A_2})P(\overline{A_3}) = 1 - (1 - 0.5) \times (1 - 0.7) \times (1 - 0.6) = 0.94$$

2. 习题二:

4.

设三等品有 k 个, 则二等品有 6k 个, 一等品有 18k 个

$$P(X = 1) = \frac{18k}{25k} = \frac{18}{25}, P(X = 2) = \frac{6k}{25k} = \frac{6}{25}, P(X = 3) = \frac{k}{25k} = \frac{1}{25}$$

X 的分布列为

X	1	2	3
P	$\frac{18}{25}$	$\frac{6}{25}$	$\frac{1}{25}$

$$\text{分布函数为 } F(x) = P(X \leq x) = \begin{cases} 0, & x < 1 \\ \frac{18}{25}, & 1 \leq x < 2 \\ \frac{24}{25}, & 2 \leq x < 3 \\ 1, & x \geq 3 \end{cases}$$

$$P(1 < X \leq 3) = P(X = 2 \cup X = 3) = P(X = 2) + P(X = 3) = \frac{7}{25}$$

$$P(1 < X \leq 3) = f(3) - f(1) = \frac{7}{25}$$

5.

由题意得, X 的可能取值为 7, 9, 13, 18;

$$P(X = 7) = \frac{C_2^2}{C_6^3} = \frac{1}{20}$$

$$P(X = 9) = \frac{C_3^2}{C_6^3} = \frac{3}{20}$$

$$P(X = 13) = \frac{C_4^2}{C_6^3} = \frac{3}{10}$$

$$P(X = 18) = \frac{C_5^2}{C_6^3} = \frac{1}{2}$$

由定义, X 的分布列为

X	7	9	13	18
P	$\frac{1}{20}$	$\frac{3}{20}$	$\frac{3}{10}$	$\frac{1}{2}$

$$\text{分布函数为 } F(x) = P(X \leq x) = \begin{cases} 0, & x < 7 \\ \frac{1}{20}, & 7 \leq x < 9 \\ \frac{1}{5}, & 9 \leq x < 13 \\ \frac{1}{2}, & 13 \leq x < 18 \\ 1, & x \geq 18 \end{cases}$$

$$\text{且有 } P(X = 7) = f(7) - \lim_{x \rightarrow 7^-} f(x) = \frac{1}{20},$$

$$P(2 < X < 7) = \lim_{x \rightarrow 7^-} f(x) - f(2) = 0,$$

$$P(7 \leq X < 13) = \lim_{x \rightarrow 13^-} f(x) - \lim_{x \rightarrow 7^-} f(x) = \frac{1}{5}$$

7.

由题意得, X 的可能取值为 1, 2, 3, 4, 5, 6

$$P(X = 1) = \frac{11}{36}$$

$$P(X = 2) = \frac{9}{36} = \frac{1}{4}$$

$$P(X = 3) = \frac{7}{36}$$

$$P(X = 4) = \frac{5}{36}$$

$$P(X = 5) = \frac{3}{36} = \frac{1}{12}$$

$$P(X = 6) = \frac{1}{36}$$

故 X 的分布列为

X	1	2	3	4	5	6
P	$\frac{11}{36}$	$\frac{1}{4}$	$\frac{7}{36}$	$\frac{5}{36}$	$\frac{1}{12}$	$\frac{1}{36}$

11.

由题意得 X 服从二项分布, X 的分布列为 $P(X = k) = C_1 5^k 0.2^k 0.8^{1-k}, k = 0, 1, \dots, 15$

$$(1) P(X = 3) = C_1 5^3 0.2^3 0.8^{1-3} \approx 0.25$$

$$4 \quad (2) P(X \geq 2) = 1 - P(X < 2) = 1 - (C_1 5^0 0.8^1 5 + C_1 5^1 0.20 0.8^1 4) \approx 0.833$$

$$(3) P(1 \leq X \leq 3) = C_1 5^1 0.20 0.8^1 4 + C_1 5^2 0.2^2 0.8^1 3 + C_1 5^3 0.2^3 0.8^1 2 = 0.613$$

$$(4) P(X > 5) = 1 - P(X \leq 5) = 1 - \sum_{k=0}^5 C_1 5^k 0.2^k 0.8^{1-k} = 0.0608$$

13.

(1) 令 A = “某人带菌”，B = “某人检测呈阳性”，X 为对某人检测三次发现阳性的次数，由题意得

$$P(X = 2) = P(X = 2|A)P(A) + P(X = 2|\bar{A})P(\bar{A}) = 0.1 \times C_3^2 0.95^2 0.05 + 0.9 \times C_3^2 0.01^2 0.99 \approx 0.0138$$

$$(2) P(A|X = 2) = \frac{P(A, X=2)}{P(X=2)} = \frac{P(X=2|A)P(A)}{P(X=2)} = \frac{0.1 \times C_3^2 0.95^2 0.05}{P(X=2)} \approx 0.9806$$

15.

令 A = ”指定的一页上至少有三个错误”

$$P(A) = 1 - P(\bar{A}) = 1 - (\sum_{k=0}^2 C_5 00^k 0.002^k (1 - 0.002)^{500-k}) \approx 1 - (\sum_{k=0}^2 \frac{1}{k!} e^{-1}) = 0.0803$$

20.

(1) 令 A = “某给定的一分钟内第一个客服未接到电话”，则 $X \sim P(2)$

$$P(A) = P(X = 0) = \frac{2^0}{0!} e^{-2} = 0.135$$

(2) 令 Y = ”5 个客服中未接到电话的人数”， $Y \sim B(5, P(A))$

$$P(Y = 4) = C_5^4 0.1353^4 (1 - 0.1353) \approx 0.0014$$

(3) 令 B_i = “5 个客服都接到 i 次电话”，C = “5 个客服接到相同电话次数”

$$\text{则 } P(X = i) = \frac{2^i}{i!} e^{-i}$$

$$P(B_i) = P(X = i)^5 = (\frac{2^i}{i!})^5 e^{-5i}$$

$$P(C) = \sum_{i=0}^{+\infty} P(B_i) = \sum_{i=0}^{+\infty} (\frac{2^i}{i!} e^{-i})^5$$

24.

$$(1) P(X = 2) = F(2) - \lim_{x \rightarrow 2^-} F(x) = 0$$

$$P(X < e) = F(e) = 1$$

$$P(2 \leq X < 3) = F(3) - F(2) = 1 - \ln 2$$

$$P(2 < X < \frac{5}{2}) = F(\frac{5}{2}) - F(2) = \ln \frac{5}{4}$$

$$(2) f(x) = F'(x) = \begin{cases} 0, & x < 1, \\ \frac{1}{x}, & 1 \leq X < e, \\ 0, & x \geq e, \end{cases}$$

27.

$$(1) \int_{-\infty}^{+\infty} f(x) dx = \int_{-1}^1 \frac{a}{\sqrt{1-x^2}} dx = a\pi = 1$$

$$\therefore a = \frac{1}{\pi}$$

$$(2) P(|X| < \frac{1}{2}) = \int_{-\frac{1}{2}}^{\frac{1}{2}} f(x) dx = \int_{-\frac{1}{2}}^{\frac{1}{2}} \frac{1}{\pi \sqrt{1-x^2}} dx = \frac{1}{3}$$

$$(3) F(x) = \int_{-\infty}^x f(x) dx = \begin{cases} 0, & x < -1, \\ \frac{1}{\pi} \arcsin x + \frac{1}{2}, & -1 \leq x \leq 1, \\ 1, & x > 1, \end{cases}$$

29.

若供电量只有 80 万 kW · h, 则供电量不能满足需求的概率为

$$P(X > 0.8) = \int_{0.8}^1 f(x) dx = 0.0272$$

若供电量只有 90 万 kW · h, 则供电量不能满足需求的概率为

$$P(X > 0.9) = \int_{0.9}^1 f(x) dx = 0.0037$$

3. 1.4-6 补充题:

先证 “A, B 相互独立 $\Rightarrow P(B|A) = P(B|\bar{A})$ ”

$\because$ A, B 相互独立

$\therefore P(AB) = P(A)P(B)$, $\bar{A}$, B 相互独立

$$P(B|A) = \frac{P(AB)}{P(A)} = \frac{P(A)P(B)}{P(A)} = P(B)$$

$$6 \quad P(B|\bar{A}) = \frac{P(\bar{A}B)}{P(\bar{A})} = \frac{P(\bar{A})P(B)}{P(\bar{A})} = P(B)$$

$$\therefore P(B|A) = P(B|\bar{A})$$

再证 “A, B 相互独立 $\Leftrightarrow P(B|A) = P(B|\bar{A})$ ”

$$\therefore P(B|A) = P(B|\bar{A})$$

$$\therefore \frac{P(AB)}{P(A)} = \frac{P(\bar{A}B)}{P(\bar{A})}$$

$$\therefore P(AB)(1 - P(A)) = P(A)(P(B) - P(AB))$$

$$\therefore P(AB) = P(A)P(B)$$

$\therefore$ A, B 相互独立

综上所述, A, B 相互独立 $\Leftrightarrow P(B|A) = P(B|\bar{A})$

2.1-6 补充题一:

由题意得, $X = 4, 5, 6, \dots$

令 $B_i =$ “前 n 次击杀怪物都没有掉落装备 A_i ”

$$\begin{aligned} P(X > n) &= P(B_1 \cup B_2 \cup B_3 \cup B_4) = P(B_1) + P(B_2) + P(B_3) + P(B_4) - \sum_{1 \leq i < j \leq 4} P(B_i B_j) + \\ &\sum_{1 \leq i < j < k \leq 4} P(B_i B_j B_k) - P(B_1 B_2 B_3 B_4) = 0.9^n \times 4 - C_4^2 \times 0.8^n + C_4^3 \times 0.7^n - 0.6^n = 4 \times 0.9^n - 6 \times 0.8^n + \\ &4 \times 0.7^n - 0.6^n \end{aligned}$$

$$\begin{aligned} P(X = n) &= P(X > n-1) - P(X > n) = 4 \times 0.9^{n-1} - 6 \times 0.8^{n-1} + 4 \times 0.7^{n-1} - 0.6^{n-1} - (4 \times 0.9^4 - 6 \times \\ &0.8^n + 4 \times 0.7^n - 0.6^n) = 0.4 \times 0.9^{n-1} - 1.2 \times 0.8^{n-1} + 1.2 \times 0.7^{n-1} - 0.4 \times 0.6^{n-1} \end{aligned}$$

2.1-6 补充题二:

由题意得

$$P(X = i, Y = j) = P(X = i, N = i + j) = P(N = i + j)P(X = i|N = i + j)$$

$$P(X = i, Y = j) = \frac{\lambda^{i+j}}{(i+j)!} e^{-\lambda} C_{i+j}^i p^i (1-p)^j = \frac{\lambda^{i+j} p^i (1-p)^{i+j}}{j! i!} e^{-\lambda}$$